



Code No.: 22CSC01

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. / B.Tech I Sem (Main) Examination March 2023

## Problem Solving and Programming

(Common to all branches)

Time: 3 Hours

Max Marks: 60

**Note:** Answer **ALL** questions from **Part-A** at one place in the same order and **Part-B** (Internal Choice)

### Part - A

(5Q X 2M = 10 Marks)

|   |                                                                    | M   | CO | BT |
|---|--------------------------------------------------------------------|-----|----|----|
| 1 | Design a flowchart for doing sum of two numbers.                   | (2) | 4  | 3  |
| 2 | Compare <i>lists</i> and <i>tuples</i> with appropriate examples.  | (2) | 3  | 4  |
| 3 | Differentiate between sequence mutation and accumulation patterns. | (2) | 1  | 2  |
| 4 | List any two advantages of modularizing program into functions.    | (2) | 3  | 1  |
| 5 | What is a file? List any two file related functions/methods.       | (2) | 6  | 1  |

### Part - B (50 Marks)

(5Q X 10M = 50 Marks)

|      |                                                                                                                                              | M   | CO | BT |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|
| 6    | (a) Summarize evolution of programming languages with appropriate examples.                                                                  | (5) | 1  | 2  |
|      | (b) What is an algorithm? Develop an algorithm for finding greatest of three numbers?                                                        | (5) | 4  | 3  |
| (OR) |                                                                                                                                              |     |    |    |
| 7    | (a) Explain program development steps.                                                                                                       | (5) | 1  | 2  |
|      | (b) Discuss software requirements for programming.                                                                                           | (5) | 1  | 2  |
| 8    | (a) What is a String? Explain any five string handling functions with suitable examples.                                                     | (5) | 3  | 2  |
|      | (b) Categorize the scope and lifetime of local and global variables in Python.                                                               | (5) | 3  | 1  |
| (OR) |                                                                                                                                              |     |    |    |
| 9    | (a) Explain about <i>arithmetic</i> and <i>relational</i> operators with appropriate examples.                                               | (5) | 2  | 2  |
|      | (b) Develop a Python program to generate prime numbers in a given range.                                                                     | (5) | 2  | 3  |
| 10   | (a) Describe iterative statements in Python with suitable program.                                                                           | (5) | 2  | 2  |
|      | (b) Write a Python program to find factorial of a given number.                                                                              | (5) | 3  | 3  |
| (OR) |                                                                                                                                              |     |    |    |
| 11   | (a) Explain the construct of <i>nested if-else</i> statement with a suitable example.                                                        | (5) | 2  | 2  |
|      | (b) Develop a python program to do matrix addition.                                                                                          | (5) | 3  | 3  |
| 12   | (a) Develop a Python program to read a word and print the number of letters, vowels and percentage of vowels in the word using a dictionary. | (5) | 5  | 3  |
|      | (b) Develop a Python program to generate Fibonacci series using a recursive function.                                                        | (5) | 3  | 3  |

(OR)

- 13 (a) What is recursion? Develop a Python script to find the Greatest Common Divisor (GCD) of given two numbers using recursion. (5) 3 1,3  
(b) Explain various dictionary methods with suitable examples. (5) 5 2
- 14 (a) How to create a text file and write content in to the text file? Explain with a python script. (5) 6 2  
(b) Explain about Python memory management operations. (5) 6 2
- (OR)**
- 15 (a) Develop a python program that reads the characters from a file and displays those characters on the screen in capital form. (5) 6 3  
(Example the file name is a '*sample.txt*' and its contents are "*Cbit Hyd*", then the output should be "*CBIT HYD*")  
(b) Explain different modes to open a file with suitable examples. (5) 6 2

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